

Spectral Krylov pretraining for quantum ground states

A baseline audit and an unconfirmed transfer hypothesis

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Abstract

Short Lanczos trajectories offer inexpensive targets for learning representations of discretized Hamiltonians. We specify a joint-embedding predictive architecture that transfers a potential encoder to ground-state prediction. The available 16-by-16-grid, single-seed experiment does not establish a benefit from Krylov pretraining. A new audit on the same retained test examples shows that a fixed free-box wavefunction attains higher mean fidelity than every retained neural method at twenty labels. Linear regression and a nine-mode Rayleigh–Ritz model perform better still. Physics residuals expose errors that high fidelity conceals. Fourteen supplementary variants pass a bounded execution check, but their tiny verification runs are not efficacy evidence. The original larger experiment and full mechanism controls remain unexecuted. The contribution at this stage is an explicit method, reproducible failure analysis and a falsifiable evaluation plan.

Introduction

Learning a map from a spatial potential to its lowest-energy wavefunction may amortize the cost of repeated eigenvalue solves. Whether pretraining reduces the number of labeled eigenpairs required is a separate empirical question. Spectral Krylov-JEPA proposes using cheap Hamiltonian actions to construct self-supervised targets before any labeled fine-tuning. The hypothesis is that representations learned from these trajectories transfer better than representations learned from potential masking or one operator action. The present evidence cannot establish that hypothesis. We therefore report the negative baseline comparison rather than an unsupported claim of improved sample efficiency.

Related work

Assran et al. [1] predict image-block representations using a joint-embedding predictive architecture. The Field-JEPA implemented here instead predicts a full-field CLS embedding from masked potential patches; it is a domain adaptation, not a canonical I-JEPA reproduction. Lanczos iteration and reorthogonalization are classical numerical methods [2]. Their algebraic properties do not prove that a neural encoder learns useful downstream representations. Physics-informed neural quantum eigensolvers [3] already learn eigenfunctions without paired training labels. Fourier neural operators [4] learn families of PDE solution maps. Neither of these broader model classes was benchmarked here; superiority to them is not claimed. The literature search and remaining novelty review are recorded in `LITERATURE_VERIFICATION.md`.

Problem formulation

Let the interior grid have n points on each axis of the unit square, spacing $h=1/(n+1)$, and zero Dirichlet boundary values. The real symmetric finite-difference Hamiltonian is $H(V) = -\Delta_h/2 + \text{diag}(V)$, with the five-point Laplacian. Ground truth is the smallest algebraic eigenpair (E_0, ψ_0) , computed with a sparse

symmetric eigensolver and residual rejection. Physical normalization is $h^2 \sum_i \psi_i^2 = 1$. Lanczos vectors instead use Euclidean unit norm. These normalizations must not be conflated. The learned map takes V and outputs an energy and a physically normalized wavefunction, with sign ambiguity handled by squared overlap.

Method and exact algorithm

Choose a reproducibly seeded unit Gaussian vector q_0 . For $j=0,\dots,K-1$, form $w=Hq_j - \beta_j q_{j-1}$, with the initial previous term zero; set $\alpha_j = q_{j-1}^T w$; subtract $\alpha_j q_{j-1}$ and fully reorthogonalize w against $q_0, \dots, q_j$; then set $\beta_{j+1} = \|w\|$ and $q_{j+1} = w/\beta_{j+1}$. Breakdown triggers a recorded deterministic restart. The corpus stores $q_0, \dots, q_K$ and the recurrence coefficients. No ground state solve is needed for pretraining data.

The potential encoder E_V and state encoder E_q tokenize spatial fields with 4-by-4 patches, learned positional embeddings and transformer blocks. The default width is 128 with four encoder blocks, four attention heads and MLP ratio four. The historical smoke preset uses width 64, two blocks and MLP ratio two. The Krylov predictor fuses potential tokens with k context states using type and step embeddings and two fusion blocks. It predicts the latent of q_k from $(V, q_0, \dots, q_{k-1})$. Corpus depth $K=3$ and prediction context $k=2$ are distinct. The target state encoder is an exponential moving average (EMA) of the online state encoder, with no target gradient. The default EMA coefficient is 0.996 and updates occur every optimizer step.

The pretraining loss is mean squared error between layer-normalized predicted and target latents. An optional head predicts α_{k-1} and β_k . Fine-tuning transfers only E_V into a shared energy head and token-aware wavefunction decoder. The frozen downstream loss combines one minus fidelity, sign-aligned mean square wavefunction error with weight 0.1, and standardized energy mean square error with weight 1. Potential and energy statistics come only from the selected labeled training subset. Validation selects the checkpoint using $(1-F)+0.5$ relative energy error. AdamW and cosine warmup schedules are used. Field, Operator and Krylov have the same downstream architecture but different pretraining parameter and token budgets. Equal step counts alone do not establish equal compute.

Mathematical analysis

Proposition 1 (classical span identity). In exact arithmetic, for a real symmetric matrix H and a nonzero q_0 , if no breakdown occurs through step k , $\text{span}(q_0, \dots, q_k)$ equals $\text{span}(q_0, Hq_0, \dots, H^k q_0)$.

Proof. The base case is immediate. Assume the identity through j . The recurrence and reorthogonalization subtract only vectors in the existing span from Hq_j , so q_{j+1} lies in the power span through $j+1$. Conversely, q_j is a polynomial of degree j in H applied to q_0 whose leading coefficient is nonzero: each nonbreakdown normalization divides by a positive β . Thus the change of basis between these vectors and the powers is triangular with nonzero diagonal, and hence invertible. This proves the identity by induction. This is a statement about the numerical target space, not a theorem about learned representations.

Proposition 2 (fidelity does not control residual without a spectral scale). Let u_0 and u_1 be orthonormal eigenvectors of H with eigenvalues E_0 and E_1 . For $\psi = \sqrt{1-\epsilon}u_0 + \sqrt{\epsilon}u_1$, its fidelity to u_0 is $1-\epsilon$, while $\|H\psi - E_0\psi\| = \sqrt{\epsilon}|E_1 - E_0|$.

Proof. Orthonormality gives the overlap directly. Subtracting $E_0\psi$ cancels the u_0 term, leaving $\sqrt{\epsilon}(E_1 - E_0)u_1$. Taking its norm proves the claim. A small infidelity may therefore coexist with a substantial residual, especially on fine grids with large high-frequency eigenvalues. No transfer theorem follows.

Proposition 3 (classical Ritz bound). For an orthonormal basis B and symmetric H , $\lambda_{\min}(B^T H B) \geq \lambda_{\min}(H)$. Enlarging a nested subspace cannot increase the minimum Ritz value.

Proof. The Rayleigh quotient minimized over vectors Bc with $\|c\|=1$ is minimized over a subset of all unit

vectors. Restricting the feasible set cannot reduce the minimum. Enlarging that feasible set cannot increase it. Our one-mode and nine-mode sine controls use nested subspaces, with numerical tests of the bound.

Experimental setup and provenance

The historical improved smoke corpus contains 40 training, 12 validation and 12 in-distribution test examples, with 8 examples per OOD family. This actual layout differs from the smaller smoke layout in the original protocol. All new baseline comparisons use its saved manifest and exact stored potentials. Fine-tuning subsets contain 10 or 20 nested labels. The historical run uses seed 11, grid 16 and 80 pretraining steps; it is a development run, not a confirmatory experiment.

Potentials are generated Gaussian mixtures, not experimental observations or molecular data. ID mixtures have one to three negative wells with amplitudes in $[-8, -2]$ and widths in $[0.08, 0.18]$. OOD families narrow the wells, deepen their amplitudes, or enforce separated double wells. Code is under the repository MIT license. No external dataset is redistributed. Seed ranges, manifests and hashes support data provenance; hashes do not establish real-world applicability.

The frozen main design is $n=32$, 1000 unlabeled potentials, training/validation/ID test counts 120/40/40, forty examples per OOD family, and nested label counts 10/25/50/100. Its runner retains SSL seed zero and downstream seeds 11/23/47. That design measures downstream-seed variation conditional on one pre-trained encoder. The supplemental full control runner independently varies pretraining and downstream seed together. Neither full campaign has executed in this work.

Benchmarks and baseline audit

The retained neural comparisons are Scratch, Field-JEPA, Operator-JEPA and Krylov-JEPA. The new controls use the analytic free-box ground-state shape, its potential-specific Rayleigh energy, and a nine-mode discrete sine Rayleigh–Ritz projection. These methods use the query Hamiltonian at inference. They are accuracy baselines with different computational costs, not claims about matched neural inference latency. A training-mean model and a linear ridge map from flattened potential to aligned wavefunction and energy use only each training subset. Ridge weight is fixed at one, with no test tuning. Predictions are normalized before measuring fidelity.

Measurements and analysis

Primary fidelity is $|\langle \psi_{\text{hat}} | \psi_0 \rangle|^2$. Energy error is $|E_{\text{hat}} - E_0| / (|E_0| + 1e-6)$. The quantity named `residual_rel` in the code is $||H \psi_{\text{hat}} - E_{\text{hat}} \psi_{\text{hat}}|| / ||\psi_{\text{hat}}||$; it is normalized by wavefunction norm, not by Hamiltonian norm, and has energy units. Residuals at the true and Rayleigh energies separately reveal wavefunction error. All paired bootstrap comparisons align sample IDs explicitly, with 2000 potential resamples. Their intervals are conditional on fitted models and exclude training-seed uncertainty. Multiple comparisons are exploratory and unadjusted; no significance-based winner is selected.

Main results

The retained Krylov model at twenty labels has ID fidelity 0.991650 and residual at the true energy 40.190890. Scratch at twenty labels has fidelity 0.993945 and residual 29.043635. The new free-box baseline has fidelity 0.994460 and residual 1.483805. Nine-mode Ritz reaches fidelity 0.999898 and residual 0.761666. Linear ridge with twenty labels reaches fidelity 0.999904 and residual 0.516766. These results show that the observed high neural fidelities do not establish a useful Krylov benefit on this development distribution. They do not falsify every possible Krylov architecture or the unrun larger study.

Under the strong-amplitude shift, the retained Krylov model has fidelity 0.965578 and relative energy error 1.151349. The Ritz baseline has fidelity 0.998979 and relative energy error 0.032930. Relative errors become

sensitive when true energies approach zero; report residuals alongside them. All audited baseline rows, including weaker training-mean energy predictions, are retained in the generated results.

Ablations and stability

Fourteen variants cover the baseline methods, context lengths one through three, shuffled trajectories, removed potential tokens, a coefficient auxiliary loss, EMA update frequency, normalized power trajectories, final latent normalization, pretraining steps and embedding width. Their 8-grid, four-step-pretrain, two-epoch-fine-tune execution checks completed every test split. This establishes operational coverage only. Coefficient loss scale and width controls are specified in `CONTROL_PROTOCOL_2026-09-27.md`. The normalized-powers control shares the K2 neural architecture and tensor dimensions, isolating orthogonalization more closely than comparisons with a smaller field-only model. It does not equalize data-generation cost. Removing V also prevents training the transferred potential encoder, a known interpretation limitation. No efficacy conclusion is drawn from this tiny matrix.

Discussion and failure modes

The development distribution is close enough to the free-box solution that high fidelity is easy to obtain without representation learning. A smooth linear map already exploits much of the potential dependence. The wavefunction decoder can introduce high-frequency errors that contribute little to overlap yet dominate physics residuals. The evidence therefore favors evaluating simple baselines and physical error before scaling training. Changing the distribution or decoder after seeing these outcomes would be a new exploratory study, requiring a separately frozen confirmation protocol.

Limitations and conclusion

The available scientific comparison is single-seed, synthetic, low resolution and post-hoc. Pretraining collapse is not ruled out by low latent prediction loss; representation-quality and full-scale multi-seed sensitivity remain incomplete. No independent reproduction, production certification, quantum advantage, foundation-model capability or state-of-the-art claim is made. Full-scale campaigns are blocked by shared local compute contention in this execution. The project is not submission-ready. The useful current result is a reproducible baseline failure and a runnable mechanism study, with preserved negative findings.

Reproducibility and AI assistance

Exact commands are in `./REPRODUCE.md`; retained receipts are at workspace root under `astra/2026-09-27/`. Tables are generated from preserved metrics by `scripts/13_audit_baselines.py`; control analysis uses `scripts/14_analyze_controls.py`. The frozen source identity is saved before each new run. Local data and checkpoints remain available but are excluded from ordinary Git storage; small receipts and hashes are retained. The repository environment snapshot records the actual versions. Codex (GPT-6 family; exact serving revision unavailable) assisted with code audit, implementation, analysis and writing. The user prompt and disclosure are included in `submission/AI_PROMPT_DISCLOSURE.md`. Human authorship, review, eligibility and consent remain unresolved; no artificial author or affiliation is supplied.

References

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